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AI Watch Adoption of Autonomous Machines

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Executive summary

This report provides an empirical analysis of the drivers and barriers to adoption of autonomous machines (AM) technologies by European companies. Based on this analysis, the report provides a series of policy recommendations to accelerate the uptake of AM and robots and emphasise its impact on the European economy.

Policy context

This report is part of AI Watch, the European Commission's knowledge service to monitor the development, uptake and impact of artificial intelligence (AI) in Europe. In December 2018, the European Commission and the Member States published a "Coordinated Plan on Artificial Intelligence" on the development of AI in the EU that mentions AI Watch's role in providing analysis and studies to support the plan. Moreover, the 2021 review of the Coordinated Plan on AI confirmed the role of AI Watch to support implementation and monitoring of the Coordinated Plan. In this context, and with an economic focus, this report contributes to two of the objectives of AI Watch: monitoring the uptake and technical developments of AI and monitoring its impact.

Key conclusions

The main takeaway of this report is that AM adoption is driven by several factors and has heterogeneous effects on companies depending on their characteristics.

Regarding the drivers of adoption, we find that firm size, employee knowledge of AI, the joint adoption of AM with complementary technologies (namely: robotic process automation, process optimisation technologies, computer vision, forecasting technologies, and creative and innovative activities) and having a low productivity level increase a firm's probability of adopting AM.

On the other hand, concerning barriers to adoption of AM, we can highlight three main findings. First, the most relevant barriers to AM adoption for non-adopters are cost of adoption and, to a lesser extent, lack of skills and data access. Specifically for large firms (those with 250 employees or more), liability and reputation risks, as well as data access, are the most important obstacles. Second, certain types of obstacles (namely: liability and reputation risks, data access and lack of funding) are more likely to be present in certain sectors of activity. Third, the more complementary technologies a firm adopts, the lower its probability of facing obstacles to AM adoption.

Policy recommendations

Based on the findings of the empirical analysis carried out, this report makes a series of policy recommendations along five axes:

- Focus on workforce skilling and upskilling
- Widen the scope of adoption-fostering policies to technology bundles
- Target firms in low-adoption sectors
- Prioritise reducing the cost of adoption
- Design policies that are tailored to the size of the company

1. Introduction

Artificial intelligence (AI)² has become an area of strategic importance with the potential to be a key driver of economic development. AI also has a wide range of potential social implications. As part of its Digital Single Market Strategy, the European Commission put forward in April 2018 a European strategy on AI in its Communication “Artificial Intelligence for Europe”. The aims of the European AI strategy announced in the communication are threefold:

- To boost the EU’s technological and industrial capacity and AI uptake across the economy, both by the private and public sectors
- To prepare for the socio-economic changes brought about by AI
- To ensure an appropriate ethical and legal framework

In December 2018, the European Commission and the Member States published a “Coordinated Plan on Artificial Intelligence” on the development of AI in the EU. The Coordinated Plan mentions the supporting role of AI Watch,³ the European Commission’s knowledge service, to monitor the development, uptake and impact of AI for Europe, which was launched in December 2018. AI Watch monitors: (i) the European Union’s industrial, technological and research capacity with respect to AI; (ii) AI-related policy initiatives in the Member States; (iii) uptake and technical developments of AI; and (iv) the impact of AI. AI Watch has a European focus within the global landscape. In the context of AI Watch, the Commission works in coordination with Member States. From AI Watch’s in-depth analyses, we will be able to understand better the European Union’s areas of strength and areas where investment is needed. AI Watch will provide an independent assessment of the impacts and benefits of AI on growth, jobs, education and society.

Subsequently, in February 2020, the Commission unveiled its vision for a digital transformation, which strives to be beneficial to the entirety of European citizens. The Commission presented a White Paper proposing a framework for trustworthy AI based on excellence and trust.

In April 2021, the European Commission proposed a set of actions to boost excellence in AI, and rules to ensure that the technology is trustworthy. The proposed Regulation on a European Approach for Artificial Intelligence and the update of the Coordinated Plan on AI, including specific actions on robotics, aim to guarantee the safety and fundamental rights of people and businesses, while strengthening investment and innovation across EU countries. The 2021 review of the Coordinated Plan on AI⁴ refers to AI Watch reports and confirms the role of AI Watch to support implementation and monitoring of the Coordinated Plan.

In this context, and as part of AI Watch, this report provides an empirical analysis of the drivers of and barriers to adoption of a key AI technologies by European companies: autonomous machines (AM)⁵ (including robots⁶). Based on this analysis, the report provides a series of policy recommendations to accelerate the uptake of AM and reinforce its impact on the European economy.

The remainder of this report is structured as follows. **Section 2** summarises the main findings of the literature regarding the three areas on which this report focuses: (i) drivers of technology adoption and (ii) barriers to technology adoption **Section 3** uses recent data from a European Commission survey to empirically analyse these three issues with respect to the particular case of AM. **Section 4** concludes and sets out policy recommendations to accelerate the uptake of AM and reinforce its impact on the European economy.

² We define artificial intelligence as “a software that is developed with one or more techniques and approaches” such as machine learning or knowledge- and logic-based approaches “and can, for a given set of human-defined objectives, generate outputs such as content, predictions, recommendations, or decisions influencing the environments they interact with” (European Commission, 2021). For a full list of these techniques and approaches, see Annex I in European Commission (2021).

³ AI Watch results and analyses are published on the [AI Watch Portal](#).

⁴ See <https://digital-strategy.ec.europa.eu/en/library/coordinated-plan-artificial-intelligence-2021-review>

⁵ We define autonomous machines as apparatus whose core function is to replicate in the most efficient way some very repetitive tasks.

⁶ We define “robots” as machines that are designed and created to operate under conditions in which human interaction is (at least) possible.

2. Factors influencing technology adoption: a literature review

The adoption of a technology such as AM is a complex process encompassing all the facets of a firm, from its infrastructure to its organisational structure and workforce skills.

Autonomous machines are self-governing machines that operate independently of direct human control. These machines are programmed to interact with their environment and make decisions without direct human intervention. Examples of autonomous machines include robots, self-driving vehicles and drones. Although research on AM has been part of the agenda of roboticists and AI scholars for some time, there has not been a unified definition of what “autonomy” means in artificial agents. Due to the existence of different definitions, it becomes difficult to assess or judge the degree of autonomy of most disruptive and sophisticated systems such as self-driving cars, industrial robots or game playing agents (Ezenkwu and Starkey, 2019). In any case, autonomous machines are extremely complex systems that integrate many pieces of technology and are still, for the most part, in the development phase and face many technical and non-technical challenges (Liu and Gaudiot, 2022).

As the data is issued from a survey concerning the use of technologies based on AI, hence the autonomous machines in object belong to the realm of AI-enhanced Robotics (Righi, 2022), with the following definition:

AI technological support for robotics: automation refers to an area close to robotics but with some substantial differences. While the term “robots” indicates machines that are designed and created to operate under conditions in which human interaction is (at least) possible, “automated machines” are apparatus whose core function is to replicate in the most efficient way some very repetitive tasks.

Nevertheless, the integration of AI technologies in automated machines allows the realisation of smart and partially self-regulating processes that may be very cost intensive because of their complexity and the physical resources/inputs involved. For this reason, the boundaries between “robotics” and “automation” are in many cases blurred and, depending on the initial perspective, a specific activity may be defined as more related to one instead of the other. Because of the continuity between these two areas, we can capture the part of “automation” that is closest to “robotics”.

In this section, we provide a concise outline of how the literature has discussed three issues related to technology adoption, namely: **(i) drivers of technology adoption and (ii) barriers to technology adoption** in order to provide a review of the research context underlying our empirical analysis in Section 3. The focus on these three topics responds to two criteria. First, these are well-studied phenomena that can inform our understanding of the drivers and barriers to AM adoption. Therefore, they are particularly pertinent to the objectives of AI Watch. Second, these are topics that we are able to explore empirically with the existing data.

2.1 Drivers of technology adoption

There is a large body of economics literature investigating how the adoption of new technologies is promoted by specific firm characteristics such as size and age (Kamien and Schwartz, 1982). Because of their perceived positive impact in technology adoption, these characteristics are often called “drivers” of technology adoption. In what follows, we briefly summarise relevant results with respect to **firm size, workforce knowledge and skills, joint adoption of complementary technologies, sector of activity and productivity level**. The choice of these factors responds to two criteria. First, the first three are among the most relevant ones in the literature. Second, for all of them we have reliable data to analyse the impact of relevant firm characteristics on AM adoption. Furthermore, as AM is one of the key technologies in the context of Industry 4.0, i.e. the new paradigm for industry relying progressively more on connection, digitalisation and automation of processes, we also highlight relevant results in this specific context. Then, the aforementioned drivers will be the object of our empirical analysis in Section 3.2.

Firm size The size of the company is an important factor in the effective adoption of technologies (Awa and Ojiabo, 2016; Awa et al., 2016; Puklavec et al., 2018; Thomas & Barton, 2012). Larger companies are able to afford new technological changes and take advantage of economies of scale. On the other hand, although small

and medium-sized enterprises (SMEs) might be aware of the relevance of new technologies, their adoption capacity is much lower than that of large companies. Larger companies have the resources to develop the right strategic planning, business formalisation and control systems to support the adoption process (Yu & Schweisfurth, 2020). Therefore, there are clear connections between company size and the capacity to embrace new technologies. In the case of Industry 4.0 technologies, which also include AM, large companies show higher abilities to make use of such technologies (Colotla et al., 2016). Notwithstanding these results, some sector-specific studies (Matsepe & Van der Lingen, 2022; Romero, & Martínez-Román, 2015; Newby et al, 2014) have found that small firms have the same or a higher probability of adopting ICT technologies than large firms.

Workforce knowledge and skills Workforce skills and learning capabilities are considered fundamental for operating new technologies in the context of Industry 4.0 (Ghadge et al., 2020; Ghobakhloo, 2020; Kamble et al., 2018; Tornatzky & Fleischer, 1990). Employees with the right education and learning capabilities are more likely to adapt to new innovative schemes and to adopt new technologies (Lin & Ho, 2011). On the other hand, a lack of knowledge hampers efforts to adopt technology (Gunasekaran & Ngai, 2008; Teo et al., 2007). Therefore, technological skills in the workplace allow companies to effectively manage the risks derived from technological investment. Nevertheless, the impact of workforce skills on the adoption of Industry 4.0 technologies can also depend on company size. In a study exploring technology adoption by SMEs in the manufacturing sector, Hassan et al (2017) find that, contrary to what previous studies showed, employee knowledge does not promote adoption in the case of SMEs.

Joint adoption of complementary technologies Technologies are characterised by their complementary nature (Gossé et al., 2020). For instance, in the manufacturing sector, process optimisation could be enhanced and synergistically build on other AI technologies such as computer vision and anomaly detection. Hence, the propensity to adopt a new technology exhibits path dependence since it is influenced by the technologies already adopted and their complementarity with the new one to be adopted. Hence, the concept of **technological bundles**, i.e. groups of complementary technologies more likely to be adopted together, has emerged (Miller et al., 2017; 2019; and Lambert et al., 2015). These results illustrate the importance of technological bundles in the innovation process. Their analysis of the agriculture sector reveals that farmers managing large operational processes requiring precise agriculture techniques have a greater propensity to adopt technology bundles. Furthermore, farmers who engage in innovation may adjust their use of a technology as new complementary techniques become available. This reveals the importance of technology bundles in innovation.

2.2 Barriers to technology adoption

The literature shows that adoption of new technologies can be hindered by obstacles such as legislative gaps, lack of data access or a weak infrastructure that prevents companies from fully exploiting their capabilities for digital transformation. In the context of our analysis, we focus on **the perception of such barriers and the factors influencing it** (Baldwin and Lin, 2002; Baldwin and Hanel, 2003; Galia and Legros, 2004; D'Este et al. 2012; D'Este et al. 2014; Pellegrino, G., 2018), which will be further empirically investigated in Section 3.3.

Innovators and non-innovators The perception of barriers to innovation is, interestingly, mostly reported by firms that are highly innovative. Most empirical studies on the perception of barriers stress a positive association between firms' perception of obstacles and their innovation propensity (Arundel, 1997; Mohnen and Rosa, 2002; Baldwin and Lin, 2002; Mohnen and Röller, 2005; Iammarino et al., 2009; Coad et al., 2016). However, as relatively recent studies suggest (Savignac, 2008, Blanchard et al. 2012; Pellegrino and Savona 2013), the positive correlation between innovators and barriers may be due to an inadequate selection of the sample: as two groups exist among non-adopters, potential innovators and firms not interested in innovation, the analysis on the perception of barriers should be restricted to the potential innovators, as they are likely to face them while adopting the technology. In this context, the concept of **revealed** and **deterring barriers** emerges (D'Este et al., 2012), based on whether the difficulties are encountered by a firm as a result and in the process of technology adoption (the former), or instead are obstacles that firms perceive as impediments to adoption (the latter).

Technological bundle effects We have seen above that AM tends to be adopted along with other complementary technologies in bundles. Therefore, when analysing barriers to AM adoption, it is pertinent to focus on technology bundles that include AM, rather than on AM alone. The literature indicates that the adoption of technological bundles can lead to stronger perception of barriers, due to the difficulties entailed by combining more than one technology (D'Este et al., 2012; de-Oliveira and Rodil-Marzábal, 2019). Previous research did not, however, disentangle the impact that certain technological bundles can play in the perception of barriers. On the other hand, the analysis we undertake in Section 3.3 sheds light on this point by investigating several technological bundles composed by AM and other complementary technologies.

2.3 The links between technology adoption and firm performance

The emergence of new and more sophisticated technologies such as AM has enhanced the impact of innovative processes on firm productivity. According to Crews (2019) and the Boston Consulting Group (CIGREF, 2018), in order to improve productivity and be more competitive, organisations will have to rely more and more on advanced digital technologies such as AI. Thus, digital transformation has an enormous potential for business value. In this section we focus on two angles: **productivity levels** and **size**. We will further explore them empirically in Section 3.4.

Productivity levels The use of new technologies disproportionately raises productivity in those firms that were already more productive (Ballestar et al., 2020; Stiebale et al., 2020). Specifically, Stiebale et al. (2020) report a considerable rise in total factor productivity (TFP) for the 20% of firms with the highest initial productivity after the adoption of industrial robots. In a related study, Dinlersoz and Wolf (2018) find that larger and more productive plants tend to be more automated. However, while robotisation⁷ positively affects mark-ups⁸ in large firms, small companies with initially low mark-ups suffer losses when adopting robots. Stiebale et al (2020) interpret that the adoption of robots seems to trigger a winner-takes-most dynamic in which large companies expand their mark-ups at the expenses of non-adopters and, to a lesser extent, of small firms adopting robots.

Firm size and productivity Regarding firm size, its effects on technology adoption are not always straightforward to predict. Economies of scale and adoption capacities influence the way firms exploit innovative processes. Some evidence indicates that both small and large firms experience productivity growth associated with the adoption of technologies, but large firms are likely to experience a more persistent growth than small firms. The main argument behind this conclusion is that large firms can more easily benefit from complementarities in the adoption of new technologies compared to small firms (Rochina-Barrachina et al., 2010).

⁷ "Robotisation" refers to the adoption of robots in general, which may not necessarily be AI-enhanced as in the present case.

⁸ Mark-ups are defined the ratio between a product's price and its cost. The adoption of robots can lead to higher mark-ups both by reducing the cost of production and by allowing to produce more sophisticated products at a higher price.

3. Empirical analysis

In this section we delve into the empirical investigation of the three factors influencing technology adoption discussed in Section 2, providing an analysis of:

1. The drivers influencing the probability of firms' adoption of AM;
2. Firms' perception of barriers to adoption;

3.1 Data

Our data are issued from a European Commission survey entitled "European enterprise survey on the use of technologies based on artificial intelligence". The survey was designed and conducted in the EU27, Norway, Iceland and the UK, resulting in the participation of 9 640 companies. It was carried out between January and March 2020. The survey focuses on AI awareness (i.e. whether the company knows what AI is), present and future AI adoption (i.e. whether the company has adopted or plans to adopt AI technologies), AI acquisition path (i.e. how does the company acquire AI: through a purchased software, developing one in-house, etc.) as well as external (e.g. the need for new laws or regulations, lack of public or external funding, etc.) and internal (e.g. lack of skills, lack of internal data, etc.) obstacles to the adoption of those technologies (for further details, see Annex A).

The sample selection was made to be representative of the EU27, Norway, Iceland and the UK.⁹ However, it is not devoid of the biases inherent in surveys, for instance the results depend on the accuracy of the responses declared by the respondents. Therefore, errors due to respondents misunderstanding a question (for example, considering they have adopted an AI-based software when in fact they have adopted a deterministic one) or failing to provide an accurate answer on behalf of the company (for example, the respondent declaring that access to data is the most relevant barrier, when most of the personnel involved with AI do not see it as such) might bias the results. In the context of surveys targeting the uptake of technologies, it is thus an effective mitigating strategy to address the survey also to the technical personnel within the company, rather than only the management. Lastly, some questions can only be answered through the subjective appreciations of the respondent. As a result, "this could mean that different respondents may have interpreted the question differently, which may have an impact on the reported figures" (IPSOS, 2019). To limit the influence of subjective appreciation, several surveys on technology adoption reformulate the questions by asking whether or not specific actions – considered as milestones for the uptake of a given technology – are undertaken in order to anchor the assessment to a concrete proxy rather than the estimation of the subject.

3.2 Adoption of AM technologies

In this section we analyse the probability of adoption of AM in EU companies, investigating the differences deriving from firm characteristics and context. Furthermore, we explore the possible relationships between the adoption of AM and the adoption of other AI technologies.

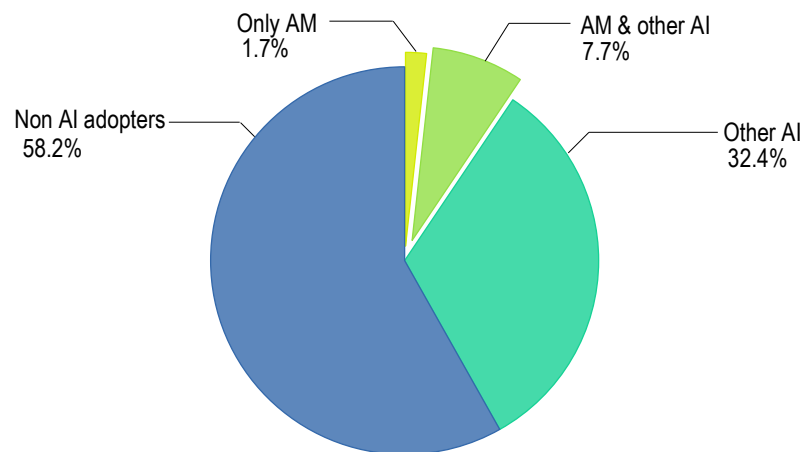
It should be noted that we can only observe whether the firm declares having adopted a given technology (be it AM or another AI technology), but not the intensity of adoption (e.g. how many AM per employee it has).

3.2.1 Probability of adoption of AM technologies

In Europe, the proportion of AM adopters is low when compared with non-AI adopters. Figure 1 illustrates the proportion of firms adopting AM in Europe, whether alone or in combination with other AI technologies. The other technologies surveyed are the following: natural language processing, computer vision, anomaly detection, sentiment analysis, forecasting, process optimisation, recommendation engines, robotic process automation (RPA), and creative activities.

⁹ For the sake of succinctness, we use the term 'Europe' to refer to these countries hereafter.

Figure 1. Breakdown of AI technology adopters and non-adopters in the data



Only 9.42% of European firms adopt AM, either alone or in combination with another AI technology, while non-AI adopters account for 58.16% of the respondents.

Thus, in order to shed light on the factors influencing such imbalance between adopters and non-adopters, we investigate the effect of three variables on the probability of AM adoption using a probit regression (for the technical details, please refer to Annex B). The variables of our interest are:

- Employees' knowledge about AI technologies,
- Firm size,
- Sector of activity,
- Firm productivity

AI knowledge Firms whose employees report having knowledge about AI technologies are more likely to adopt autonomous machines and robots compared to firms whose employees report a lack of knowledge of AI technologies. We can hence confirm that, according to our results (cf. Figure 2 below), and in coherence with the literature (cf. Section 2.1), employees with the right competences and knowledge about AI technologies favour adoption of AM and robots.¹⁰ Note that the survey from which these results are drawn only contains data about AI knowledge in general, as opposed to AM knowledge in particular. This is explained by two intertwined reasons. First, while our analysis focuses on AM, the survey's objective is to understand the uptake and effects of several AI technologies. Second, employee knowledge of AI in general favours the uptake of several AI technologies, as it provides a knowledge base on which individuals can develop specific AI-related knowledge. Consequently, as shown in Section 3.2.2, AM adoption is linked to the adoption of other complementary technologies that benefit from the same general AI knowledge.

Firm size As outlined in the previous subsection, SMEs can also face difficulties in the adoption of AM due to weaker competitive advantages compared to larger firms. Absence of economies of scale, along with a limited financial capacity, complicate AM adoption by small size companies. This, in turn, could entail uncertainty regarding amortisation (Birkel et al., 2019).

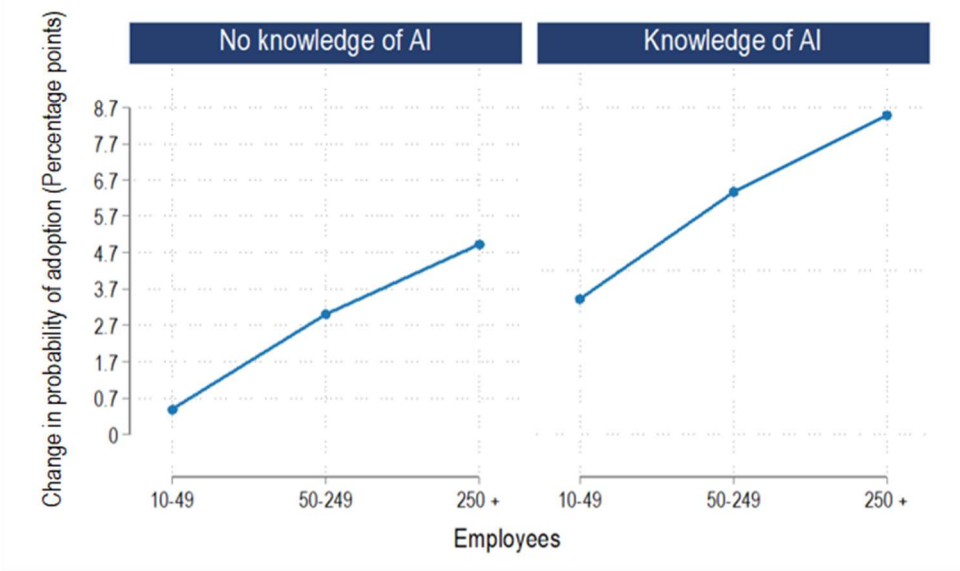
Figure 2 illustrates the effects of company size on the probability of AM adoption. The plot shows the marginal effects¹¹ of size over AM adoption probability. To assess the influence of companies having employees with the right competencies, we split the analysis into firms whose employees are knowledgeable about AI technologies

¹⁰ Note that these employees could have acquired knowledge about AI before the adoption of AM or been hired before the adoption.

¹¹ The marginal effect gives the effect on the probability of adoption while holding all other variables constant.

and firms whose employees are not. The full results are also reported in Table S1 in the “Additional tables” section.

Figure 2. Change in the probability of adoption of AM technologies by firm size and employee knowledge of AI (compared to micro firms’ probability)



Firms whose employees already have knowledge of AI technologies show a higher probability of adopting AM compared to those whose employees do not. Both graphs report a positive and relevant effect on the capacity of medium and large-sized companies to adopt AM and robots as compared to small firms. The analysis indicates that a firm with a size of 50–249 employees has a probability of adopting AM, which is 3.2 percentage points higher than a micro-firm of 5–9 employees. This difference increases to 5.4 percentage points when we compare with large firms (250+ employees). Hence, larger companies show clear advantages over SMEs. In this context, companies that enjoy higher economies of scale are able to adopt and exploit those technologies more easily than smaller companies with less resources. This may create a tighter competitive environment that leaves many small companies struggling to find ways to optimise their productive processes.

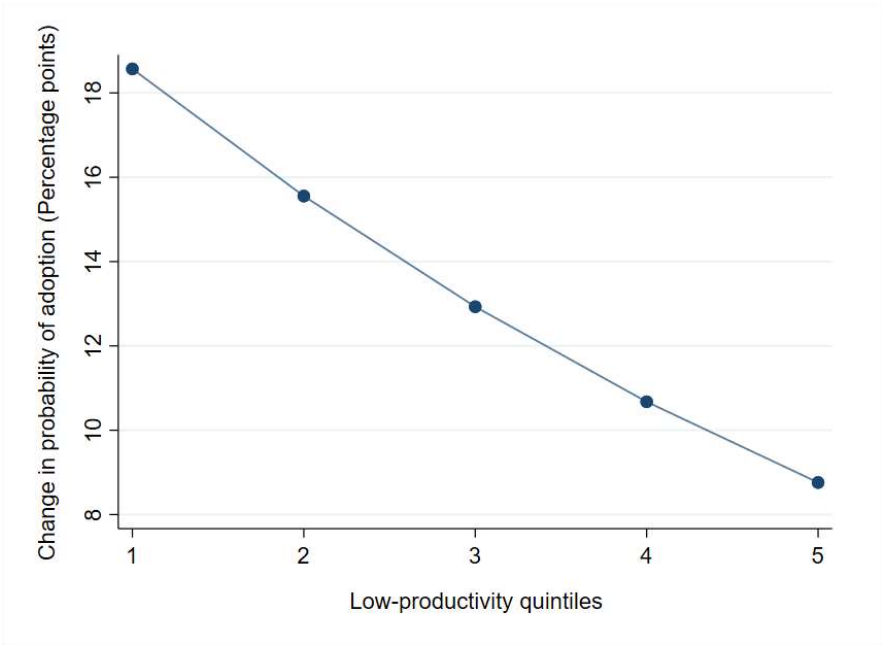
Sector of activity Figure 3 below shows the change in probability of adoption of AM technologies across sectors (the sectorial breakdown of the survey data can be found in Table A1). In this figure (as well as in those that follow, which differentiate results across sectors) we use the agriculture sector as a benchmark against which to assess how certain sectors can affect the probability of adoption of AM.¹²

We also find strong differences in AM adoption by sector. Figure 3 displays the fluctuation of probability of AM adoption across sectors. Companies operating in the construction, trade and transport, IT and finance, and education/human health sectors show lower probabilities of adoption than companies operating in agricultural activities. Firms working in the trade and transport sector show the lowest likelihood of adoption compared to firms competing in the agricultural sector. Concretely, a firm operating in the trade and transport sector has a probability of adopting AM technologies 12 percentage points lower than a firm engaged in agricultural activities. The risk of harm to persons that the activities of the transportation sector entail, and the consequent regulatory requirements to introduce AI-driven solutions, might explain this. On the other hand, many studies delve into the disruption brought about by robotic and autonomous processes in the agricultural sector, mainly by large-sized companies. Higher population projections and the effects of climate change put pressure on

¹² The choice of a sector as a benchmark is necessary when analysing the impact of a given sector on the probability of the outcome (in this case, adopting AM) using a probit model. Any sector can be chosen as a benchmark without altering the results, as the model evaluates relative changes in probabilities between sectors (i.e. how much more likely it is to adopt AM for a firm in sector A compared to a firm in sector B), as opposed to absolute probabilities (what is the average probability of adoption of AM by firms belonging to sectors A versus firms belonging to sector B).

relationship exists for low-productivity firms¹³. In particular, after segmenting the sample into 5 quintiles of productivity, and controlling for all other firm characteristics, we can see that there is a negative relationship between a firm’s productivity and its probability of adopting AM technologies for low-productivity (i.e., the 20% less productive firms) firms. Figure 4 below illustrates this finding.

Figure 4 Change in probability of adoption of AM technologies by productivity level for the 20% less productive firms



NB: low-productivity firms are the 20% (i.e., first quintile) less productive firms of the entire sample.

We believe that the negative relationship between firm-level productivity and AM adoption for low-productivity firms can be attributed to the efforts of the latter to catch up with their higher-productivity counterparts. The adoption of AM technologies may present an opportunity for these firms to improve their productivity levels. As the literature shows (cf. Section 2.3), the adoption of robots by a firm boosts its productivity.

3.2.2 Joint adoption of complementary technologies

As mentioned above, in the survey, firms reported the adoption of several AI technologies apart from AM. Table 1 shows the proportion of firms adopting AM in combination with any of the other nine AI technologies. The technologies that present the highest percentages of firms having also adopted AM are robotic process automation (RPA) (40.9%) and process optimisation (35.9%). This confirms the importance of considering the existence of complementarities between technologies. In line with the studies developed by Miller et al. (2017) and Lambert et al. (2015), technology adoption is contingent upon to the adoption of other technologies that are complementary to it.

¹³ See table S7 and table S8. Adoption of AM technologies may depend on a firm’s level of productivity, while productivity itself may be affected by prior adoption of these technologies. This would require correcting for the presence of potential endogeneity bias in the estimation. Unfortunately, we were unable to correct for this bias due to the unavailability of appropriate instruments. Nonetheless, we believe that the presence of endogeneity is unlikely to alter the nature of our findings.

Table 1. AI technologies adopted with AM

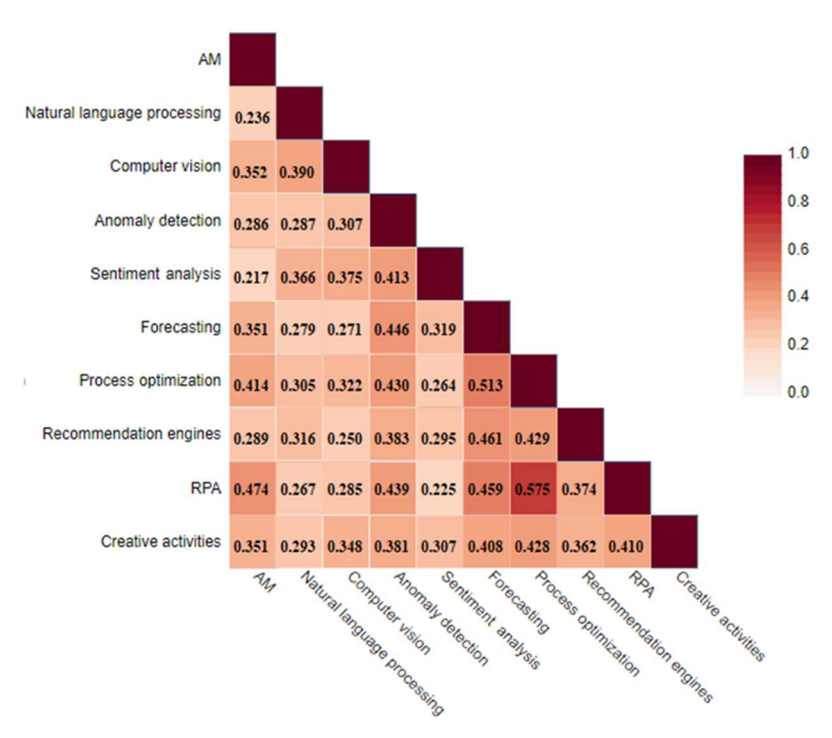
Technologies	N° of firms	%
Natural language processing	166	20.34
Computer vision	213	26.10
Anomaly detection	229	28.06
Sentimental analysis	56	6.86
Forecasting	210	25.74
Process optimization	293	35.91
Recommendation engines	171	20.95
RPA	334	40.93
Creative activities	160	19.61

Thus, we study the relationship between technologies and the probability of joint adoption through a Multivariate Probit Model (MPM) for the ten categories of technologies described in the survey (for further details on the methodology, please refer to Annex B).

Error! Reference source not found. provides the error correlation matrix¹⁴ of the MPM between the ten technologies. In the figure, the depth of colour designates the magnitude of the correlation: the deeper the colour, the stronger the correlation between the technologies. In this case, all ten categories are positively correlated (in red) to each other. This indicates that all AI technologies mutually reinforce each other at the firm level.

Figure 5. Correlation matrix between technologies

¹⁴ The correlation between the errors of the 10 equations estimating the probability of adopting each of the 10 technologies studied provide a measurement of the extent to which the probability of adopting technology A depends on the probability of adopting technology B. Here, the term “error” refers to the part of the choice of the probability of choosing a certain technology that cannot be explained by the variables introduced in the model, such as firm size or sector of activity. If the errors of the equations estimating the probabilities of adopting technologies A and B are correlated, this indicates that there is some unobservable variable not present in the dataset (e.g. the fact that the two technologies are complementary in that their joint use is necessary to introduce a certain productive process) that explains that, to a certain extent, the choice of adopting technologies A and B are linked.



We have used a two-step approach to statistically determine which technologies appear to be most used together with AM. In the first step, adoption patterns were analysed using a MPM model. Next, the model offers an error correlation matrix for technology adoption (Figure 5). Using these correlation values, we are able to isolate clusters of technologies that are frequently used together. To cluster the technologies into bundles, we use a threshold correlation. In order to identify the existence of a bundle, the correlation between AM and other technologies must be above the mean correlation level (0.33). Five technologies are over this threshold. Among those technologies, two stand out as outliers, as their correlation coefficient is particularly high: process optimization and RPA. Hence, we choose the lowest correlation level of the two (process optimization: 0.414) as the second threshold level. This results in two bundles. The first bundle (correlation threshold: 0.4 or more) is composed by: RPA and process optimisation technologies. The second bundle (correlation threshold between 0.33 and 0.4) is composed by computer vision, forecasting and creative activities. Figure 6 illustrates the bundle selection according to both thresholds.

Figure 6. Bundles of technologies adopted together with AM – thresholds

2012). Therefore, among a total of 5,038 non-adopters, we include in the analysis 1,528 firms who confirmed their intentions to innovate in the future. We consider those firms as potential innovators, independently of the fact that they have not adopted any technology yet.

Based on the questionnaire design, we classify the perception of barriers to adoption by considering five categories: (1) data access; (2) liability and reputation; (3) cost; (4) skills; and (5) internal data. In the analysis of their impact, we again focus on the adoption of the two technology bundles. Tables 2 and 3 show the ratio of firms reporting each of the aforementioned barriers as important. The tables compare two sub-samples: adopters, who face revealed barriers, and non-adopters, who in turn face deterring barriers, as defined in Section 2.2. For each variable, a means comparison test showed that the difference between the aforementioned ratios are statistically significant at the 0.01 level.

Table 2. Ratio of firms reporting obstacles to adoption (Bundle 1)

	Total	Deterring	Revealed
Data access	0.72	0.77	0.69
Liability and reputation	0.64	0.69	0.61
Cost adoption	0.89	0.93	0.87
Lack of skills	0.73	0.79	0.69
Lack of internal data	0.45	0.51	0.41

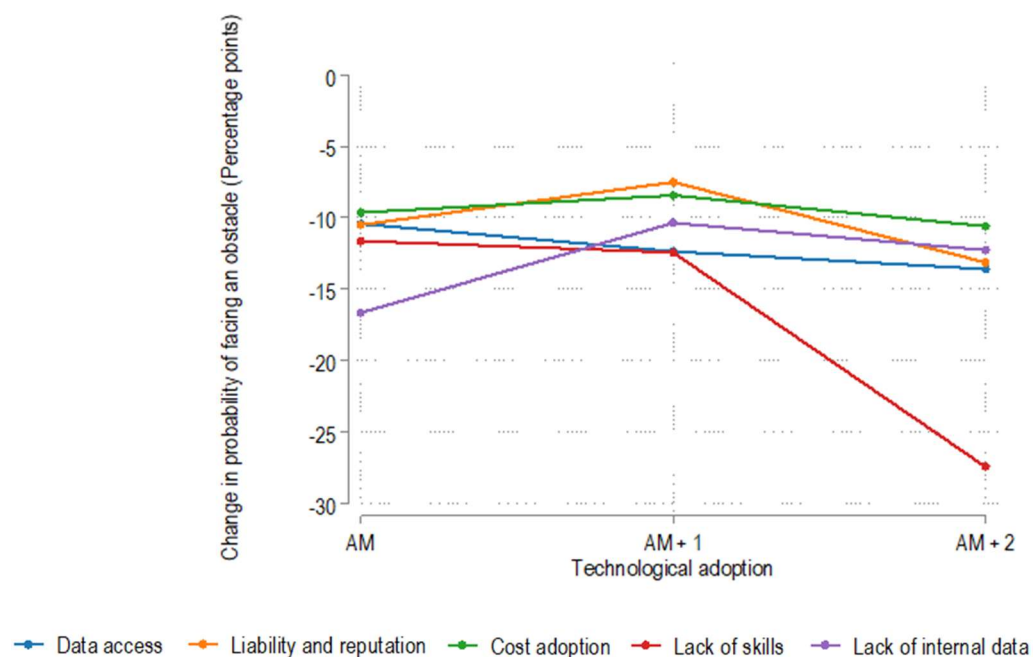
Table 3. Ratio of firms reporting obstacles to adoption (Bundle 2)

	Total	Deterring	Revealed
Data access	0.73	0.78	0.70
Liability and reputation	0.65	0.71	0.62
Cost adoption	0.89	0.93	0.87
Lack of skills	0.73	0.81	0.67
Lack of internal data	0.45	0.49	0.43

In both bundles, “cost of adoption” is the category among all obstacles to adoption that firms consider as the most important one or with a higher impact over their decisions. This was already the case when considering all of the technology categories. On the other hand, and despite the considerable number of firms reporting it, the lack of internal data seems to be a comparatively minor barrier for innovators and potential innovators when compared to the other obstacles reported. Both tables also show that differences exist in perceived barriers between adopters and potential adopters. For instance, deterring barriers appear to be more frequent than revealed ones. A possible source of this discrepancy may come from a substantial difference between the two cohorts, as adopters, having successfully deployed a given technology, have most likely overcome a series of obstacles which still stand as possibility for potential adopters. Hence the importance of distinguishing the influence of the different types of barriers, in line with the literature (D’Este et al., 2012; de-Oliveira and Rodil-Marzábal, 2019; Pellegrino, G., 2017).

We obtained the probability of facing an obstacle through a multivariate ordered probit model for the two technology bundles (for the complete results, refer to Tables S3 and S6 in the additional tables section). Contrary to previous results (D’Este et al. 2012; Pellegrino and Savona 2013), our analysis highlights a complex and non-linear relationship between obstacle perception and technology adoption, linked to the type of barrier. Figure 7 displays the likelihood of facing obstacles to adoption as firms adopt more and more technologies of Bundle 1, while Figure 8 displays the probability of facing obstacles to adoption as firms adopt more technologies of Bundle 2.

Figure 7. Change in the probability of facing an obstacle to adoption (compared to non-adopters' probability) for different adoption levels (Bundle 1)



Note: The three cohorts in the graph are: AM: the adoption of only AM technologies; AM+1: the adoption of AM with one technology among RPA or process optimization; and AM+2: the entire bundle. All are compared to the non-adopters of the bundle. Only statistically significant obstacles to adoption depicted.

Figure 8. Change in the probability of facing an obstacle to adoption (compared to non-adopters' probability) for different adoption levels (Bundle 2)

4. Conclusions and policy recommendations

Despite the rapid pace of technological change, the adoption rate of autonomous machines and robots in the EU is relatively low. Only close to 9.5% of EU companies claim adoption. In contrast, in 2018 the world average sector-level adoption rate was estimated to be of 17.5%. Several types of barriers hold back adoption (cost, lack of internal and external data, insufficient workforce skills and liability, reputation concerns and lack of funding), which are particularly felt by small firms. Notwithstanding such hurdles, as shown by the literature, companies that manage to adopt AM technologies benefit from productivity increases.

The empirical analysis carried out in this report allowed us to have a better understanding of the drivers and barriers to adoption of AM by European firms. The main takeaway is that **AM adoption is a multifactorial process that plays out and impacts companies differently depending on their characteristics**. Consequently, this report calls for **a holistic and targeted policy design to accelerate the adoption of AM and its economic impact**. In particular, we set out a series of policy recommendations structured around the following five axes.

Focus on workforce skilling and upskilling

Companies whose employees report knowledge of AI technologies have a higher probability of adopting AM regardless of other characteristics of the firm (size or sector). Hence, policies aiming at skilling and upskilling the workforce could act as a booster to the uptake of AM and its most complementary technologies (i.e., technological bundles); this crucial role played by training is also echoed the results of several studies with a sectorial focus, for instance, in manufacturing (de Nigris et al, 2022) and health (de Nigris et al, 2020).

More concretely, **in the short run**, the deployment of one-stop shops to support companies by providing (among other services) training and skills development such as **European Digital Innovation Hubs**¹⁵ can function as a basis for responding to this need. This can be **complemented with other firm-level (up-)skilling policies** such as incentives for employers to develop training programmes for employees with skills similar to those required for AM use; online trainings and courses to supplement individual skill gaps; and strengthening the bridges between university graduates with the skills needed to adopt AM (e.g., mathematics or computer science) and companies. The Digital Skills and Jobs¹⁶ platform and the AI4EU¹⁷ stand as examples of such endeavours. **In the long run**, policies should focus on **ensuring that university and tertiary education programmes provide the necessary competences for graduates to develop AM skills, and that they are able to attract enough students across Member States**. In this sense, programmes aiming at developing university-industry collaborations are key, such as the European Network of AI Excellence Centres.¹⁸

Widen the scope of adoption-fostering policies to technology bundles

The empirical analysis carried out in this report confirmed that, as is generally the case with other technologies, AM adoption benefits from the adoption of other complementary technologies that form “technology bundles”. **Consequently, policies aimed at increasing the uptake of AM should widen their scope to also target its complementary technologies**: RPA, process optimisation technologies, computer vision, forecasting technologies, and creative and innovative activities. Therefore, firm-level policies could benefit from assessing in which AM-complementary technologies a company is lagging behind, and focus on their adoption together with AM.

Target firms in low-adoption sectors

¹⁵ See <https://digital-strategy.ec.europa.eu/en/activities/edihs>

¹⁶ <https://digital-skills-jobs.europa.eu/en>

¹⁷ <https://www.ai4europe.eu/>

¹⁸ <https://www.elise-ai.eu/>

Our analysis shows that barriers to AM adoption (be it alone or together with other complementary technologies) vary across sectors. Trade/Transport, IT/Finance and Construction are three sectors that appear as particularly low adopters.

Therefore, **policymakers could target** firms located in the three aforementioned sectors, and **in particular Trade/Transport and IT/Finance, as they present the lowest AM adoption levels.**

Prioritise reducing the cost of AM adoption

Not all barriers to adopting AM and its complementary technologies are equally important. Our results show that the cost of adoption stands out as the most relevant one. Therefore, policies should focus on alleviating the economic burden of adopting AM and its complementary technologies faced by companies. Policies such as tax breaks for adopters and public co-financing innovation projects involving AM and its complementary technologies can help lift this barrier. Lack of internal skills and data access, on the other hand, appear as the second most relevant obstacles to AM adoption. In this regard, policies aimed at fostering data sharing in line with the Data Act¹⁹ should continue to be developed. This includes, but is not limited to, the development of Common European Data Spaces²⁰ as well as sectoral data-sharing policies that supplement the provisions of the Data Act.

Design policies that are tailored to company size

Our analysis shows that small companies are the type of firms that adopt AM the least. Therefore, **policy design should focus on prioritising smaller firms and tailoring its instruments to their specificities.** The fact that these companies present the highest potential to boost AM adoption calls for avoiding a one-size-fits-all AM uptake policy.

In the case of large firms (firms with at least 250 employees), liability- and reputation-related obstacles appear as the most relevant obstacles to AM adoption. In that respect, policies aiming at providing legal certainty in terms of the application of liability rules to AI-driven products and services should foster the adoption of AI technologies by large firms with the highest capacity to bring these technologies to the common European market. **The ongoing adaptation and harmonisation of civil liability rules and the Product Liability Directive** to the specific challenges of AI are encouraging steps in that direction.

¹⁹ See <https://digital-strategy.ec.europa.eu/en/policies/data-act>

²⁰ See <http://dataspaces.info/common-european-data-spaces/#page-content>